Four finite rows in the classification of string C-groups

Human Compression project

We recover the rows for S_{12}, S_{13}, S_{14} in Table 1 and for A_{15} in Table 3 of the fixed survey [1]. The common enumeration, intersection criterion and stable high-rank theory are acquired results. The contribution here is the complete execution and verification of the remaining finite cases. The source tables specify the questions, rather than supply their answers as premises.

Theorem 1. *The following are the complete counts in the indicated ranks. An ordered class means simultaneous point conjugacy of an ordered marking. A self-dual class is fixed by reversal. The last column counts the underlying generating sets under group automorphisms, as in the source.*

Group	*Rank*	*Ordered classes*	*Self-dual classes*	*Classes with duality*
S_{12}	*3*	*6567*	*125*	*3346*
	4	*1875*	*5*	*940*
	5	*364*	*2*	*183*
	6	*150*	*0*	*75*
	7	*74*	*6*	*40*
S_{13}	*3*	*14181*	*145*	*7163*
	4	*1723*	*3*	*863*
	5	*341*	*1*	*171*
	6	*244*	*2*	*123*
	7	*82*	*0*	*41*
S_{14}	*3*	*45891*	*361*	*23126*
	4	*7869*	*21*	*3945*
	5	*1951*	*5*	*978*
	6	*606*	*0*	*303*
	7	*324*	*2*	*163*
	8	*108*	*0*	*54*
A_{15}	*3*	*11557*	*83*	*5820*
	4	*314*	*2*	*158*
	5	*40*	*0*	*20*
	6	*81*	*3*	*42*
	7	*12*	*0*	*6*

The remaining nonzero counts for S_{12}, in ranks $8, 9, 10, 11$, are $9, 7, 1, 1$. For S_{13}, in ranks $8, 9, 10, 11, 12$, they are $35, 9, 7, 1, 1$; for S_{14}, in ranks $9, 10, 11, 12, 13$, they are likewise $35, 9, 7, 1, 1$. All higher ranks have count zero, and A_{15} has no marking of rank at least eight.

Complete enumeration and independent verification

Proof. We apply the acquired permutation enumeration [2], leaving its mathematical engine unchanged. In the three symmetric groups, every first involution is conjugate to one having e disjoint transpositions, with $1 \le e \le \lfloor n/2 \rfloor$. In A_{15} the allowable values are $e = 2, 4, 6$. We classify by the full automorphism group of A_{15}, namely S_{15} acting by point conjugacy, so these are exactly the three initial classes needed for that group as well.

The acquired proof visits each permissible next involution through its orbit under the actual pointwise centralizer of the prefix. It pays both the intersection test on the growing prefix and the transitivity pruning. Every one of the resulting twenty-two initial branches terminates: six branches in each of degrees twelve and thirteen, seven in degree fourteen, and three for A_{15}. The rank limits are respectively $7, 7, 8, 7$. Thus its completeness theorem applies throughout these ranges; a timeout is not interpreted as exhaustion. Filtering by the actual generated group leaves $9030, 16571, 56749, 12004$ tuples, respectively. The full transcripts and the exact engine used in each branch are retained.

The search excludes a commuting adjacent pair for these targets by the acquired indecomposability argument. Indeed, such a pair separates a string C-marking into commuting nontrivial factors whose intersection is trivial by the intersection property. These symmetric groups have no such direct-product decomposition, and A_{15} is simple. Hence the noncommutation graph is the connected path on the marked generators. An isomorphism of their underlying generating sets must preserve that path. It either preserves their order or reverses it. All automorphisms of the symmetric groups here are inner, and all automorphisms of A_{15} are induced by S_{15}. Consequently point conjugacy and reversal give precisely the source equivalence relation.

We verify all positive tuples independently of the search state. The verifier reconstructs each literal permutation, checks nonidentity, involution orders and the distant commutations, and computes its actual generated subgroup. It requires equality with the stated target group. For every contiguous interval $[a, b]$ of at least three marks it checks

$$\langle t_a, \dots, t_{b-1} \rangle \cap \langle t_{a+1}, \dots, t_b \rangle = \langle t_{a+1}, \dots, t_{b-1} \rangle.$$

The acquired interval theorem [3] proves that these equalities, with its rank-one and rank-two bases, imply the full intersection property. These are equalities of actual permutation subgroups. They do not rely on cached prefix results or on projection of an intersection property. The four lists require $16986, 25066, 94392, 13729$ interval comparisons: 150173 comparisons on 94354 complete tuples.

For distinctness and reversal we use the acquired literal-array invariant of [4]. Starting at a point, traverse its orbit in the order of the marked generators and express all generators in that resulting order of points. Minimize this array over all starting points. Transitivity ensures that every traversal covers the whole domain. A point conjugacy carries a rooted traversal to the corresponding rooted traversal, whereas equality of two arrays supplies the actual conjugating bijection. Thus two minimized arrays agree exactly when the ordered markings are point conjugate. All arrays in each list are distinct. The minimized reversed array is also located in that list; the resulting permutation of the list is checked to be an involution. Its fixed points give the self-dual column. Its number of orbits is therefore half the sum of the ordered and self-dual columns, as asserted.

The higher-rank statements use acquired proofs, rather than further search. For S_n, write $k = n - r$. The complete stable correspondence and its finite base counts [5] give constants $1, 1, 7, 9, 35$ for $k = 1, \dots, 5$, provided $n \geq 2k+3$. In degree twelve the remaining ranks have $k \leq 4$; in degrees thirteen and fourteen they have $k \leq 5$. Each meets the threshold, and substitution gives the stated tails. The retained independent-generator bound $r \leq n-1$ for S_n excludes all greater ranks. For A_{15}, the complete acquired alternating-rank theorem [6] gives $r \leq \lfloor (15-1)/2 \rfloor = 7$. This proves the theorem in every rank occurring in the two source tables. □

Certificate, provenance and scope

The twenty-two complete branch transcripts are stored with lossless compression. Their original uncompressed hashes, terminal receipts, rank headings and complete literal outputs are checked before any tuple is used. The checker reconstructs the same involution universe and positive lists from those outputs. Its default mode repeats all actual group and interval checks and every literal conjugacy/reversal check. The option `--replay-enumeration` additionally repeats the unchanged enumeration branches with their recorded individual limits and compares the complete results. The report explicitly distinguishes these modes. Interrupted executions and incomplete transcripts cannot supply a count.

The final common validator differs from the previously executed positive validators only by selecting the explicitly named target group and by its completion marker. Its canonical traversal is unchanged. The final replay compares every resulting array with the original complete positive phase, allowing only that literal marker change. It also checks the entire reversal permutation, rather than merely its number of fixed points. Damaged completion and duplicate-class controls must reject, as must optimized Python execution. The development record retains an initial extraction failure: the helper mistakenly included intervening branch records after the involution list. It stopped before a mathematical validation; the corrected delimiter changes no census transcript or mathematical engine.

The original numerical results remain attributed to the survey and its collaborators. This component supplies complete recovery and verification for twenty-one formerly unrecovered experimental cells. The high-rank entries were already acquired. No new open-conjecture result, priority, whole-original admission, Lean certificate or compression measurement is claimed. The manuscript leaves the survey's other claims to their explicitly separate proofs and source qualifications.

References

[1] D. Leemans, *String C-group representations of almost simple groups: a survey*, arXiv:1910.08843v2, Tables 1 and 3, physical pages 6 and 8.

[2] Human Compression project, *Complete permutation searches and the six-point row*, accepted component and its reviews.

[3] Human Compression project, *Checking intersection properties on intervals*, accepted component and distinct root review.

[4] Human Compression project, *The complete string C-group counts for* S_{11}, accepted component and distinct root review.

[5] Human Compression project, *Pure cuts and the stable counting correspondence*, and *Balanced pairs and the finite base counts*, acquired through the complete L019 principal recovery.

[6] Human Compression project, complete L044 principal recovery, with the original Cameron–Fernandes–Leemans–Mixer theorem and its prior lower-existence constructions explicitly attributed there.